MACHINE LEARNING FOR MOLECULAR DISCOVERY

Utku Umur Acikalin

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Fourth-year Cornell CS Ph.D. student advised by Carla P. Gomes. Builds machine learning to identify unknown molecules from tandem mass spectra: neuro-symbolic spectrum prediction, learned search over molecular graphs, and retrieval over 105M structures, with discoveries confirmed by synthesis.

EDUCATION

Cornell University · Ph.D. in Computer Science — 2023 - expected 2028
Advisor: Carla P. Gomes

TOBB University of Economics and Technology — Ankara, Turkey
M.S., Computer Engineering, 2019 - 2022 · GPA 4.00/4.00
B.S., Computer Engineering, 2015 - 2019 · GPA 3.74/4.00

RESEARCH EXPERIENCE

Ph.D. Researcher · Cornell University, Gomes Lab · with the Schroeder Lab (chemistry) — 2023 - present

AIMe & MS²KOSMOS: neuro-symbolic mass spectrometry at database scale

Co-first author · bioRxiv 2026, under review at Nature · Preprint · Public system

Outcome: synthesis-validated discovery of previously unreported microbiota-dependent polyamines.

- Co-designed and implemented **AIMe**, a neuro-symbolic MS/MS predictor that explains peaks through chemically constrained fragmentation pathways; **0.83 cosine on NIST** vs. 0.75 for the strongest baseline.
- Built CPU/GPU retrieval over **105M structures and 800M+ predicted spectra** (MS²KOSMOS), with interactive queries in a few seconds and a repository-scale search of 7.14M GNPS consensus spectra.
- Co-developed the public web system for spectral search, pathway inspection, and on-demand prediction.

GEMS: structure elucidation as learned search over isomer graphs

First author · under review at ICLR 2027 · ICML 2026 workshops: GenBio (spotlight), AI4Physics · arXiv

- Developed a learned search that refines candidates through formula- and valence-preserving bond edits; a policy maps MIST-predicted fingerprints to atoms and bonds and scores all edits in one pass.
- Designed shortest-path supervision, enabling pretraining on PubChem structures alone and fine-tuning on spectrum-predicted fingerprints.
- **Hit@1 of 29.6% on NPLIB1 and 30.2% on MassSpecGym** with the target excluded from initialization, above published baselines in comparable settings; higher accuracy than FRIGID on NPLIB1 in ~2.6 min vs. 26 h.

MoRE: Mixture of Reused Experts

Co-first author · COLM 2026 · arXiv · Code

- Co-developed a middle ground between standard MoE and looped (Universal) Transformers: adjacent layers share an expert pool with per-layer routers, giving ~18x more routing choices at no parameter cost.
- Beats a DeepSeek-v3-style MoE and MoEUT at 114M-1.15B parameters with matched parameters and FLOPs.

X²GNN: learning to explore and exploit

First author · ICLR 2025 · Code

- Unsupervised GNN solver for MIS, Max Clique, Max-Cut; trained on one refinement step, it scales with test-time compute: 1.2% from optimal on large Max Clique graphs (prior neural: >22%), beating Gurobi at similar runtime.

SELECTED PUBLICATIONS & PREPRINTS

* Equal contribution.

Intellectual property protection lost and competition: An examination using large language models.
U. U. Acikalin, T. Caskurlu, G. Hoberg, G. M. Phillips.
Journal of Financial Economics 182:104306, 2026.

Charting the small-molecule universe from mass spectra with neuro-symbolic AI.
U. U. Acikalin*, D. Feng*, A. M. Ferber*, G. J. Gouveia*, et al.
bioRxiv 2026; under review at Nature.

GEMS: Molecular Structure Elucidation as Learned Search over Isomer Graphs.
U. U. Acikalin, Y. Wang, G. J. Gouveia, T. Schwertfeger, F. C. Schroeder, C. P. Gomes.
arXiv 2026; under review at ICLR 2027. Earlier version: ICML 2026 workshops, GenBio (spotlight) and AI4Physics.

MoRE: Mixture of Reused Experts.
E. S. Qiu*, **U. U. Acikalin***, J. Lovelace, C. Belardi, A. B. Mulchandani, C. P. Gomes, K. Q. Weinberger.
COLM 2026.

Learning to Explore and Exploit with GNNs for Unsupervised Combinatorial Optimization.
U. U. Acikalin, A. M. Ferber, C. P. Gomes.
ICLR 2025.

Unsupervised Combinatorial Probabilistic Reasoning: Probabilistic Coin Change Problem.
Z. Qu, Y. Wang, **U. U. Acikalin**, et al.
AAAI 2026.

Earlier work (11 papers, 2019 - 2025): combinatorial optimization (INFORMS Journal on Computing, Constraints, GECCO), algorithmic game theory, NLP (Natural Language Engineering, SIGIR PatentSemTech workshop), and economics. Full list at utkuumur.github.io.

TECHNICAL SKILLS

Machine learning: PyTorch, Hugging Face Transformers, scikit-learn; graph neural networks, mixture-of-experts transformers, learned search policies, LLM-based text extraction.

Molecular ML: RDKit; MS/MS spectrum prediction and retrieval, fragmentation modeling, molecular fingerprints (Morgan, MIST), isomer-graph search; NIST20, MassSpecGym, NPLIB1, CASMI, GNPS.

Optimization and HPC: integer programming (Gurobi, CPLEX), OR-Tools, A* and heuristic search; CPU/GPU similarity search at 100M+ scale, MPI, Cerebras CSL, AWS.

Languages: Python, C++, Java, JavaScript.

PRESENTATIONS, SERVICE & TEACHING

ICML 2026, Seoul: GEMS at GenBio (spotlight) and AI4Physics; MoRE at Weight-Space Symmetries.

Reviewer: NeurIPS 2026; ICML 2026 GenBio Workshop; IEEE Transactions on Circuits and Systems I.

Teaching assistant: Introduction to Computer Programming; Design and Analysis of Algorithms; Social Networks; Combinatorics.

SELECTED AWARDS

TUBITAK 2210 M.S. Scholarship, 2020 - 2022; **TOBB ETU Graduate Special Merit Scholarship**, 2019 - 2021; **TUBITAK Undergraduate Research Project Contest**, 2nd prize, 2019.